

Find Gradient (slope, $m = -$)

$$y = x^3 \quad \begin{matrix} (2, 8) \\ (x, y) \end{matrix}$$

$$\frac{dy}{dx} = \frac{d}{dx}(x^3)$$

$$\frac{dy}{dx} \Big|_{x=2} = \underline{\underline{3x^2}} = 3(2)^2 = 3 \times 4 = 12$$

Gradient = 12

$$\frac{(x+3)(3x-1)}{x} = \frac{3x^2 - x + 9x - 3}{x}$$

$$= 3x + 8 - 3x^{-1}$$

$$= 3 + 0 + 3x^{-2}$$

$$= 3 + 3x^{-2}$$